# Computer Science Capstone Topic Approval Form

The purpose of this document is to help you clearly explain your capstone topic, project scope, and timeline. Identify each of these areas so that you will have a complete and realistic overview of your project. Your instructor cannot sign off on your project topic without this information.

*Note: You must fill out and submit this form. Space beneath each number will expand as needed.*

*Note: Any costs associated with developing the application will be the responsibility of the student.*

**INFORM INSTRUCTOR:**
Potential use of proprietary company information: No

**ANALYSIS:**

## 1. Project topic and description:

**Stock Price Predictor:** This project aims to develop a machine learning model capable of predicting future stock prices based on historical stock data and potentially other relevant factors.

## 2. Project purpose and goals:

The primary purpose is to create a tool that can assist investors in making informed decisions about buying or selling stocks. The goal is to achieve a model that, when used as the basis of a buy/sell strategy, outperforms the S&P 500.

## 3. Descriptive method:

**Line Chart:** This is appropriate because it seems to be the most common way to visualize stock prices over time. The x-axis would represent time, and the y-axis would represent the price. Multiple lines could be used to compare the model's predicted price to the actual stock price, or to compare the performance of a portfolio based on the model's predictions to the S&P 500.

**Candlestick Chart:** This type of chart is also frequently used in stock analysis. Each "candlestick" represents the open, high, low, and close price for a given period. This can provide more detailed information about price movements than a simple line chart.

**Bar Chart:** While less common for displaying stock prices over time, bar charts would be useful to show the daily or weekly returns of the model's predictions compared to the actual returns, or to compare the overall performance of the model to the S&P 500 over a specific period.

## 4. Predictive or prescriptive method:

**Polynomial Regression:** Since the model will be predicting a number, regression models are appropriate. Stock prices are non-linear, so polynomial regression will be appropriate for capturing cyclical market behavior.

**Residual Calculation:** Residuals are the difference between the actual stock price and the predicted stock price. This quantifies the fluctuation around the trend.

**Random Forest Classifier:** Because random forests are designed to represent the average of the results of randomized inputs, they are robust against overfitting. By using the residuals for training, the classifier focuses on predicting fluctuations that the regressor did not capture. The classifier's predictions of short-term fluctuations can be used to refine the overall price prediction by adjusting the regressor's trend prediction. Alternatively, the classifier's confidence in the predictions can be used as a signal to adjust the trading strategy, perhaps by only taking positions when the classifier's confidence is above a certain threshold.

**DESIGN and DEVELOPMENT:**

1. Computer science application type (select one):
   - ~~Mobile (indicate Apple or Android)~~
   - ~~Web~~
   - Stand-alone

2. Programming/development language(s) you will use: Python

3. Operating system(s) or platform(s) you will use: Google Colab

4. Database Management System you will use: DuckDB

5. Estimated number of hours for the following:
   i. Planning and design:10
   ii. Development:10
   iii. Documentation: 5
   iv. Total: 25

6. Projected completion date:
   Feb. 16, 2025

**IMPLEMENTATION and EVALUATION:**

1. Describe how you will approach the execution of your project.

   1. Data Collection: Import or download and load data from reliable sources, potentially including financial news and public datasets.

   2. Data Preprocessing: Clean and prepare the data for analysis, including handling missing values and normalizing data.

   3. Feature Engineering: Extract relevant features from the data that may be predictive of stock prices.

4. Model Training: Train the polynomial regressor on the prepared data to get predictions on price.

5. Residual Calculation: Subtract the predicted prices from the known prices to get the residuals, which will be used to train the random forest classifier.

6. Model Training: Train the random forest classifier on the residuals to predict the fluctuation around the regressor's model of the trend.

7. Model Evaluation: Evaluate the performance of the model using appropriate metrics, such as mean squared error or accuracy, and cross-validation.

8. Fine-Tuning: Adjust model parameters and potentially try different algorithms to improve performance. For the regressor, cross-validation and regularization will help with overfitting. For the classifier, tuning could involve parameters such as number of trees, max. depth, min. samples per leaf, etc.

- ☐ **This project does not involve human subjects research and is exempt from WGU IRB review.**

**STUDENT'S SIGNATURE**

By signing and submitting this form, you acknowledge that any costs associated with the development and execution of the application will be your (the student's) responsibility.

**INSTRUCTOR'S SIGNATURE:**

**INSTRUCTOR APPROVAL DATE:** 2.13.25